

Lagrange's Equation of Motion

The equation of motion of a vibrating system can be obtained in terms of the generalised co-ordinates using Lagrange's equation of motion. The Lagrange's equation of motion for n-degree of freedom system can be stated as,

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = Q_j^{(n)}; \quad j = 1, 2, \dots, n$$

$\dot{q}_j$ is the generalised velocity and $Q_j^{(n)}$ is the non-conservative generalised forces corresponding to the generalised co-ordinates q_j . The forces represented by $Q_j^{(n)}$ may be dissipative (damping) forces or other external forces that are not derivable from a potential function. For example, if F_{xk}, F_{yk}, F_{zk} represent the external forces acting on the k^{th} mass of the system in the x, y, and z directions, respectively, the generalised force $Q_j^{(n)}$ can be calculated as

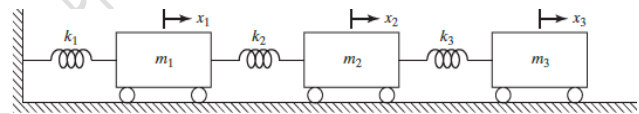
$$Q_j^{(n)} = \sum_k \left(F_{xk} \frac{\partial x_k}{\partial q_j} + F_{yk} \frac{\partial y_k}{\partial q_j} + F_{zk} \frac{\partial z_k}{\partial q_j} \right)$$

where x_k, y_k , and z_k are the displacements of the k^{th} mass of the system in the x, y, and z directions respectively. For a conservative system, $Q_j^{(n)} = 0$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = 0$$

Example 1

Determine the equations of motion of the three degree of freedom system shown using Lagrange's equation of motion



Let the generalised coordinates of the system $q_j = x_1, x_2, x_3$

Kinetic energy of the system,

$$T = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2$$

The spring k_1 deforms by x_1 , spring k_2 deforms by $(x_2 - x_1)$, and spring k_3 deforms by $(x_3 - x_2)$

Strain energy of the system,

$$V = \frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_2 - x_1)^2 + \frac{1}{2} k_3 (x_3 - x_2)^2$$

Since there are external forces or damping forces in the system, the generalised forces $Q_j^{(n)} = 0$. Therefore, the Lagrange's equation becomes

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = 0$$

$$\frac{\partial T}{\partial \dot{q}_j} = \frac{\partial}{\partial \dot{x}_j} \left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2 \right)$$

$$\frac{\partial T}{\partial q_j} = \frac{\partial}{\partial x_j} \left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2 \right)$$

$$\frac{\partial V}{\partial q_j} = \frac{\partial}{\partial x_j} \left(\frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_2 - x_1)^2 + \frac{1}{2} k_3 (x_3 - x_2)^2 \right)$$

For the first generalised coordinate $q_1 = x_1$

$$\frac{\partial}{\partial \dot{x}_1} \left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2 \right) = \frac{1}{2} m_1 (2\dot{x}_1)$$

$$\frac{\partial}{\partial x_1} \left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2 \right) = 0$$

$$\frac{\partial}{\partial x_1} \left(\frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_2 - x_1)^2 + \frac{1}{2} k_3 (x_3 - x_2)^2 \right) = \frac{1}{2} k_1 (2x_1) + \frac{1}{2} k_2 \times 2(x_2 - x_1)(-1) + 0$$

Therefore, the Lagrange's equation becomes

$$\frac{d}{dt} \left(\frac{1}{2} m_1 (2\dot{x}_1) \right) - 0 + \frac{1}{2} k_1 (2x_1) + \frac{1}{2} k_2 \times 2(x_2 - x_1)(-1) = 0$$

$$m_1 \ddot{x}_1 + (k_1 + k_2)x_1 - k_2 x_2 = 0 \text{ --- (a)}$$

For the second generalised coordinate $q_2 = x_2$

$$\frac{\partial}{\partial \dot{x}_2} \left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2 \right) = \frac{1}{2} m_2 (2\dot{x}_2)$$

$$\frac{\partial}{\partial x_2} \left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2 \right) = 0$$

$$\frac{\partial}{\partial x_2} \left(\frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_2 - x_1)^2 + \frac{1}{2} k_3 (x_3 - x_2)^2 \right) = 0 + \frac{k_2}{2} \times 2(x_2 - x_1) + \frac{k_3}{2} \times 2(x_3 - x_2)(-1)$$

Therefore, the Lagrange's equation becomes

$$\frac{d}{dt} \left(\frac{1}{2} m_2 (2\dot{x}_2) \right) - 0 + \frac{k_2}{2} \times 2(x_2 - x_1) + \frac{k_3}{2} \times 2(x_3 - x_2)(-1) = 0$$

$$m_2 \ddot{x}_2 - k_2 x_1 + (k_2 + k_3)x_2 - k_3 x_3 = 0 \text{ --- (b)}$$

For the third generalised coordinate $q_3 = x_3$

$$\frac{\partial}{\partial \dot{x}_3} \left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2 \right) = \frac{1}{2} m_3 (2\dot{x}_3)$$

$$\frac{\partial}{\partial x_3} \left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} m_3 \dot{x}_3^2 \right) = 0$$

$$\frac{\partial}{\partial x_3} \left(\frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_2 - x_1)^2 + \frac{1}{2} k_3 (x_3 - x_2)^2 \right) = \frac{1}{2} k_3 \times 2(x_3 - x_2)$$

Therefore, the Lagrange's equation becomes

$$\frac{d}{dt} \left(\frac{1}{2} m_3 (2\dot{x}_3) \right) - 0 + \frac{1}{2} k_3 \times 2(x_3 - x_2) = 0$$

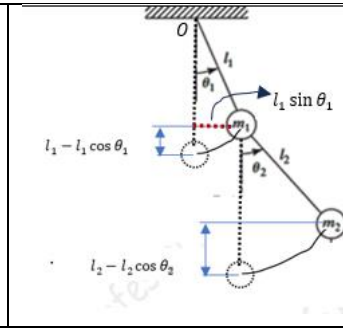
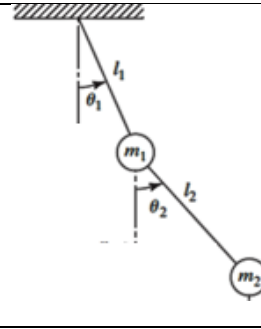
$$m_3 \ddot{x}_3 - k_3 x_2 + k_3 x_3 = 0 \text{ --- (c)}$$

Writing equations (a), (b), and (c) in matrix form,

$$\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{Bmatrix} + \begin{bmatrix} (k_1 + k_2) & -k_2 & 0 \\ -k_2 & (k_2 + k_3) & -k_3 \\ 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

Example 2

Determine the equations of motion of the for the double pendulum shown using Lagrange's equation of motion. Obtain the natural frequencies assuming $m_1 = m_2$ and $l_1 = l_2$



Considering the generalised coordinates q_j as θ_1 and θ_2

The potential energy of the system

$$V = m_1 g(l_1 - l_1 \cos \theta_1) + m_2 g\{(l_1 - l_1 \cos \theta_1) + (l_2 - l_2 \cos \theta_2)\} \quad (1)$$

Since the angles are very small, the displacement of mass m_1 , $x_1 = l_1 \theta_1$

the displacement of mass m_2 , $x_2 = l_1 \theta_1 + l_2 \theta_2$

The kinetic energy of the system

$$\begin{aligned} T &= \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 = \frac{1}{2} m_1 (l_1 \dot{\theta}_1)^2 + \frac{1}{2} m_2 (l_1 \dot{\theta}_1 + l_2 \dot{\theta}_2)^2 \\ &= \frac{1}{2} m_1 l_1^2 (\dot{\theta}_1)^2 + \frac{1}{2} m_2 l_1^2 (\dot{\theta}_1)^2 + \frac{1}{2} m_2 l_2^2 (\dot{\theta}_2)^2 + \frac{1}{2} m_2 (2l_1 l_2 \dot{\theta}_1 \dot{\theta}_2) \\ T &= \frac{1}{2} (m_1 + m_2) l_1^2 (\dot{\theta}_1)^2 + \frac{1}{2} m_2 l_2^2 (\dot{\theta}_2)^2 + m_2 (l_1 l_2 \dot{\theta}_1 \dot{\theta}_2) \quad \dots (2) \end{aligned}$$

Since there are external forces or damping forces in the system, the generalised forces $Q_j^{(n)} = 0$. Therefore, the Lagrange's equation becomes

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = 0 \quad \dots (3)$$

For the first generalised coordinate $q_1 = \theta_1$

$$\begin{aligned} \frac{\partial T}{\partial \dot{q}_1} &= \frac{\partial}{\partial \dot{\theta}_1} \left\{ \frac{1}{2} (m_1 + m_2) l_1^2 (\dot{\theta}_1)^2 + \frac{1}{2} m_2 l_2^2 (\dot{\theta}_2)^2 + m_2 (l_1 l_2 \dot{\theta}_1 \dot{\theta}_2) \right\} \\ &= \frac{1}{2} (m_1 + m_2) l_1^2 (2\dot{\theta}_1) + m_2 (l_1 l_2 \dot{\theta}_2) \end{aligned}$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_1} \right) = \frac{d}{dt} \left(\frac{1}{2} (m_1 + m_2) l_1^2 (2\dot{\theta}_1) + m_2 (l_1 l_2 \dot{\theta}_2) \right) = (m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 (l_1 l_2 \ddot{\theta}_2)$$

$$\frac{\partial}{\partial \theta_1} \left\{ \frac{1}{2} (m_1 + m_2) l_1^2 (\dot{\theta}_1)^2 + \frac{1}{2} m_2 l_2^2 (\dot{\theta}_2)^2 + m_2 (l_1 l_2 \dot{\theta}_1 \dot{\theta}_2) \right\} = 0$$

$$\frac{\partial V}{\partial q_j} = \frac{\partial}{\partial \theta_1} (m_1 g(l_1 - l_1 \cos \theta_1) + m_2 g\{(l_1 - l_1 \cos \theta_1) + (l_2 - l_2 \cos \theta_2)\}) = m_1 g l_1 \sin \theta_1 + m_2 g l_1 \sin \theta_1$$

For small angles of oscillations, $\sin \theta_1 \cong \theta_1$

$$\frac{\partial V}{\partial q_j} = (m_1 + m_2) g l_1 \theta_1$$

$$\therefore (3) \Rightarrow (m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 + (m_1 + m_2) g l_1 \theta_1 = 0 \quad \dots (a)$$

For the second generalised coordinate $q_2 = \theta_2$

$$\begin{aligned} \frac{\partial T}{\partial \dot{q}_2} &= \frac{\partial}{\partial \dot{\theta}_2} \left\{ \frac{1}{2} (m_1 + m_2) l_1^2 (\dot{\theta}_1)^2 + \frac{1}{2} m_2 l_2^2 (\dot{\theta}_2)^2 + m_2 (l_1 l_2 \dot{\theta}_1 \dot{\theta}_2) \right\} \\ &= \frac{1}{2} m_2 l_2^2 (2\dot{\theta}_2) + m_2 (l_1 l_2 \dot{\theta}_1) \end{aligned}$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_2} \right) = \frac{d}{dt} \left(\frac{1}{2} m_2 l_2^2 (2\ddot{\theta}_2) + m_2 (l_1 l_2 \ddot{\theta}_1) \right) = m_2 l_2^2 \ddot{\theta}_2 + m_2 (l_1 l_2 \ddot{\theta}_1)$$

$$\frac{\partial}{\partial \theta_2} \left\{ \frac{1}{2} (m_1 + m_2) l_1^2 (\dot{\theta}_1)^2 + \frac{1}{2} m_2 l_2^2 (\dot{\theta}_2)^2 + m_2 (l_1 l_2 \dot{\theta}_1 \dot{\theta}_2) \right\} = 0$$

$$\cdot \frac{\partial V}{\partial q_j} = \frac{\partial}{\partial \theta_2} (m_1 g (l_1 - l_1 \cos \theta_1) + m_2 g \{ (l_1 - l_1 \cos \theta_1) + (l_2 - l_2 \cos \theta_2) \}) = m_2 g l_2 \sin \theta_2$$

For small angles of oscillations, $\sin \theta_2 \cong \theta_2$

$$\frac{\partial V}{\partial q_j} = m_2 g l_2 \theta_2$$

$$\therefore (3) \Rightarrow m_2 l_2^2 \ddot{\theta}_2 + m_2 (l_1 l_2 \ddot{\theta}_1) + m_2 g l_2 \theta_2 = 0$$

$$m_2 l_1 l_2 \ddot{\theta}_1 + m_2 l_2^2 \ddot{\theta}_2 + m_2 g l_2 \theta_2 = 0 \text{ --- (b)}$$

Writing equations (a) and (b) in the matrix form

$$\begin{bmatrix} (m_1 + m_2) l_1^2 & m_2 l_1 l_2 \\ m_2 l_1 l_2 & m_2 l_2^2 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} (m_1 + m_2) g l_1 & 0 \\ 0 & m_2 g l_2 \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \text{ --- (c)}$$

Taking $m_1 = m_2 = m$ and $l_1 = l_2 = l$, equation (c) becomes

$$\begin{bmatrix} 2ml^2 & ml^2 \\ ml^2 & ml^2 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} 2mgl & 0 \\ 0 & mgl \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

This is now of the form $[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$

Assuming a harmonic solution

$$x_1(t) = X_1 \sin(\omega t + \phi)$$

$$x_2(t) = X_2 \sin(\omega t + \phi)$$

This can be written as $\{x\} = \{X\} \sin(\omega t + \phi)$, where $\{x\} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$ and $\{X\} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix}$

Differentiating the above equation two times, we get

$$\{\ddot{x}\} = -\omega^2 \{X\} \sin(\omega t + \phi)$$

Therefore, the matrix equation now becomes

$$-\omega^2 [M] \{X\} \sin(\omega t + \phi) + [K] \{X\} \sin(\omega t + \phi) = \{0\}$$

$$([K] - \omega^2 [M]) \{X\} \sin(\omega t + \phi) = \{0\}$$

$$([K] - \omega^2 [M]) \{X\} = \{0\} \text{ --- (d)}$$

For this system of linear homogeneous equations to have a non-trivial solution,

$$\det\{[K] - \omega^2 [M]\} = 0$$

$$\begin{vmatrix} 2mgl - \omega^2 2ml^2 & -\omega^2 ml^2 \\ -\omega^2 ml^2 & mgl - \omega^2 ml^2 \end{vmatrix} = 0$$

$$2(mgl - \omega^2 ml^2)^2 - \omega^4 m^2 l^4 = 0$$

$$2m^2 g^2 l^2 - 4\omega^2 m^2 g l^3 + 2\omega^4 m^2 l^4 - \omega^4 m^2 l^4 = 0$$

$$l^2 \cdot \omega^4 - 4gl \cdot \omega^2 + 2g^2 = 0 \text{ --- (e)}$$

Solving the above quadratic equation in ω^2

$$\omega^2 = \frac{4gl \pm \sqrt{(4gl)^2 - 4 \cdot l^2 \cdot 2g^2}}{2 \cdot l^2} = \frac{4gl \pm \sqrt{8g^2 l^2}}{2l^2} = \frac{4gl \pm \sqrt{8}gl}{2l^2} = \left(\frac{4 \pm \sqrt{8}}{2} \right) \frac{g}{l} = 0.59 \frac{g}{l}, 3.41 \frac{g}{l}$$

$$\omega = 0.77 \sqrt{\frac{g}{l}}, 1.85 \sqrt{\frac{g}{l}}$$

When $\omega = \omega_1 = 0.77\sqrt{\frac{g}{l}}$, Eqn (d) becomes

$$\begin{bmatrix} 2mgl - \omega_1^2 2ml^2 & -\omega_1^2 ml^2 \\ -\omega_1^2 ml^2 & mgl - \omega_1^2 ml^2 \end{bmatrix} \begin{Bmatrix} X_1^1 \\ X_2^1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$0.814mgl X_1^1 - 0.593mgl X_2^1 = 0$$

$$\frac{X_2^1}{X_1^1} = \frac{0.814}{0.593} = 1.373$$

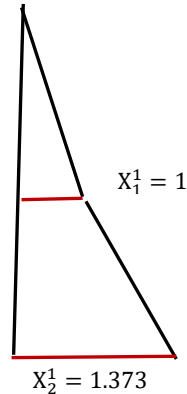
When $\omega = \omega_2 = 1.85\sqrt{\frac{g}{l}}$, Eqn (d) becomes

$$\begin{bmatrix} 2mgl - \omega_2^2 2ml^2 & -\omega_2^2 ml^2 \\ -\omega_2^2 ml^2 & mgl - \omega_2^2 ml^2 \end{bmatrix} \begin{Bmatrix} X_1^2 \\ X_2^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$-4.845mgl X_1^2 - 3.422mgl X_2^2 = 0$$

$$\frac{X_2^2}{X_1^2} = \frac{-4.845}{3.422} = -1.416$$

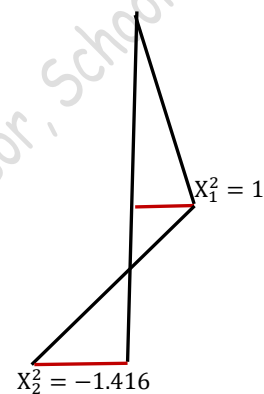
I mode shape



Normal modes or modal vectors

$$\{X\}^1 = \begin{Bmatrix} X_1^1 \\ X_2^1 \end{Bmatrix} = X_1^1 \begin{Bmatrix} 1 \\ 1.373 \end{Bmatrix}$$

II mode shape

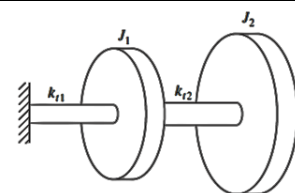


Normal modes or modal vectors

$$\{X\}^2 = \begin{Bmatrix} X_1^2 \\ X_2^2 \end{Bmatrix} = X_1^2 \begin{Bmatrix} 1 \\ -1.416 \end{Bmatrix}$$

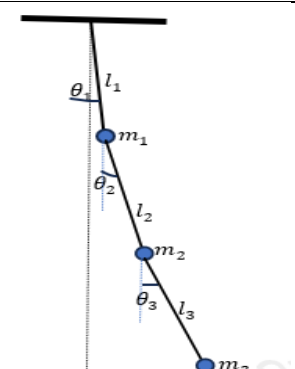
Exercise 1

Using Lagrange's equations, derive the differential equations for the small amplitudes of oscillations for the two-rotor system shown assuming $J_2 = 2J_1$ and $k_{t2} = 2k_{t1}$



Exercise 2

Using Lagrange's equations, derive the differential equations for the small amplitudes of oscillations for a triple pendulum.



Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition